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HANOI UNIVERSITY
OF SCIENCE AND TECHNOLOGY

INTRODUCTION TO BUSINESS ANALYTICS

Capstone Project 2026 · Topic 7

Profit-Aware Telecom Churn and Retention Targeting

Given a limited retention budget, which customers should receive an offer when both churn risk and expected economic value are considered?

Final Project Report · Group 1

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Analysis run `20261029T034118630540Z-88be4b93` · every number in this report is generated from that run

Hanoi, October 29, 2026

Executive Summary

Decision. A telecom operator has a fixed retention budget and must decide, every campaign wave, *which customers receive a retention offer and which offer they receive*. The current practice is to call the customers most likely to leave. This report asks whether adding each customer's economic value and the cost of an offer to the decision produces more value from the same budget, and how much should be spent.

Approach. Following CRISP-DM, we framed the decision first (unit: one customer; outcome: churn; decision: offer or not, and which offer, within a budget), then built a calibrated churn model on the IBM Telco Customer Churn sample (7,043 customers, 26.5% churners), a transparent customer-value proxy, and an expected-value (EV) decision rule

$$\mathrm{EV}_{i,o} \;=\; p_i \times q_o \times V_i \;-\; (c_o^{\text{offer}} + c^{\text{contact}})$$

where p_i is the predicted churn probability, q_o the assumed *save rate* of offer o, and V_i the customer's discounted 12-month margin. An integer program chooses customers and offers within the budget. All choices were made on a validation split; the held-out test split (1,409 customers) was used once.

Findings.

- **The model ranks churners well and its probabilities can be trusted for money calculations:** calibrated logistic regression, ROC-AUC 0.830, PR-AUC 0.620 (random baseline 0.265), expected calibration error 0.019. Random forest and XGBoost did not rank better, so the simpler, explainable model is deployed.
- **Same budget, more value.** With \$3,500 on the test customers, *Expected value + segment offers* yields an expected net value of \$2,668 (ROI 76%) against \$1,301 for today's highest-risk rule: **+\$1,368**, of which \$1,039 comes from choosing customers by expected value and \$328 from matching offers to segments. Valued with customers' actual churn the gain is \$1,077 (95% bootstrap interval \$229 to \$2,002).
- **Who gets an offer:** 100% of the recommended list have fibre internet and 97% a month-to-month contract, with a median bill of \$89 (\$70 overall). Calling the most *valuable* customers loses money (−\$1,037), and so does random contact (−\$1,645): risk and value are needed together.
- **How much to spend:** keeping only budget steps that still return at least 10% gives about \$45,737 for the whole base (\$6.49 per customer); on the test customers this buys 324 offers for \$9,150 with expected net value \$4,145.
- **Main risk:** the save rate is an assumption, not a measurement. The recommended list breaks even if the standard offer saves 13.8% of would-be leavers (assumed 20%) and loses money in 17% of 5,000 Monte Carlo draws of all assumptions; expected-value targeting nevertheless beats risk-only targeting in 99.96% of draws. *Who* to target is robust; *how much* an offer saves is not.

Recommendation. Adopt expected-value targeting with segment-specific offers, and launch it as a **randomised 90-day pilot** of 1,620 customers (half held out as a control). Scale up only if the measured save rate of the offer mix is statistically significant and at least 15.6%, its break-even (assumed: 22.6%). The data contain no randomised campaign, so no causal uplift is claimed; all money figures are expected values under visible, documented assumptions in USD of the fictional sample company.

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1 Introduction

Customer churn is the central economic problem of subscription businesses: acquiring a replacement customer usually costs more than keeping an existing one, and every departure removes a stream of future margin. Most churn projects stop at predicting *who will leave*. A retention manager, however, does not need a ranking but a *decision*: whom to contact, with which offer, and how much to spend, given that every offer costs money and not every customer is worth saving [3, 4].

This project answers Topic 7 of the course capstone brief:

> **Business question.** Given a limited retention budget, which customers should receive an offer when both churn risk and expected economic value are considered?

We follow the CRISP-DM process taught in the course (Figure 1). Each phase produced an artefact in the repository, and the sections of this report follow the same order. The work is fully reproducible: one command (`python -m telco_retention run-all`) downloads the data with a checksum, trains, evaluates, simulates and regenerates every table, figure, notebook, document and this report.

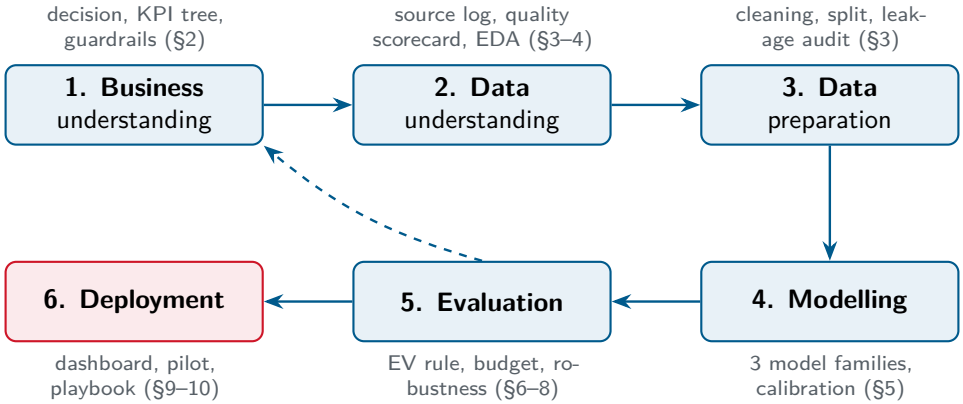


Figure 1: CRISP-DM roadmap of the project and where each phase is reported. The dashed arrow marks the loop back to the business question after evaluation.

Contributions. Beyond the required baseline (logistic regression, a tree model, calibration, threshold analysis and an EV simulator compared with simple rules), the project adds: (i) a cross-validated feature review that removes a collinear input; (ii) an exact budget-constrained optimiser with an offer catalogue and a contact cap; (iii) a profit curve that sizes the budget on validation data; (iv) one-way, two-way, scenario and Monte Carlo sensitivity analysis, including break-even save rates; (v) a stress test for lower churn and Vietnamese price levels; (vi) a fairness audit of who receives offers; (vii) a statistically powered pilot design; and (viii) an interactive decision dashboard.

2 Business Understanding

2.1 Problem statement: decision, options, criteria

Following the Decision–Options–Criteria template from the course, the problem is framed around the decision before any model is built (Table 1).

Table 1: Problem framing.

Element	Definition in this project
Context	A subscription telco loses about one in four customers in the sample period; the retention team has a fixed budget per campaign wave and a call centre with limited capacity.
Decision owner	Head of Customer Retention (with Finance for costs and margin, the call centre for capacity, Legal/DPO for consent).
Decision	For each campaign wave: the contact list, the offer per customer, and the size of the budget.
Options	(a) no campaign; (b) call the highest-risk customers (current practice); (c) call the most valuable customers; (d) random contact; (e) rank by expected value with one standard offer; (f) expected value with segment-specific offers.
Criteria (KPIs)	Expected net retention value and ROI (primary); precision among contacted; coverage of high-risk/high-value customers; customers contacted.
Guardrails	At most 30% of customers contacted; selection-rate ratio between groups within 0.8–1.25 or reviewed; no use of age or gender as model inputs; consent and do-not-call checks.
Unit of analysis	One customer (one row of the data).
Target	`Churn` = customer left during the observation window (binary).
Decision time and horizon	The scoring snapshot; the outcome of an offer is read after a 90-day window; customer value covers the next 12 months (scaled by tenure).

2.2 KPI tree

The North-Star metric of a campaign wave is its *expected net retention value*. Decomposing it multiplicatively (Figure 2) shows the levers the business controls and which of them the analysis can inform: *who* is contacted (the model and the decision rule), *which offer* (the catalogue), and *how many* (the budget). The save rate is the least known driver; it is a lagging indicator that can only be measured by an experiment.

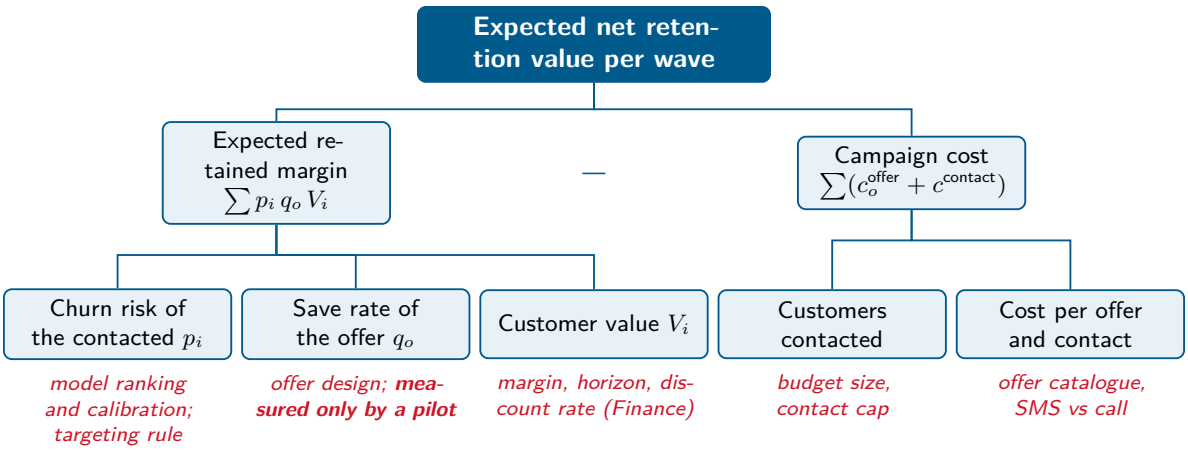


Figure 2: KPI tree of a retention campaign wave. Red text: the lever that moves each driver.

Leading and lagging indicators. Precision among contacted customers and coverage of high-risk/high-value customers are *leading* indicators available as soon as a list is drawn. The realised save rate and net value per customer are *lagging*: they appear only after the outcome window, by comparing offered customers with a randomised hold-out group.

2.3 Success criteria, constraints and assumptions

- **Technical success:** PR-AUC and ROC-AUC clearly above the prior-only baseline on held-out data, with good calibration (expected calibration error below 0.05 and calibration slope near 1), because expected value multiplies a probability by money.
- **Business success:** at the same budget, the recommended rule must deliver a higher expected net value than the current highest-risk rule, and the advantage must survive the sensitivity analysis.
- **Constraints:** fixed budget per wave (base case: enough to give the standard offer to 10% of customers, i.e. $3,500 for the 1,409 test customers); one offer per customer; offer eligibility rules; contact cap of 30%.
- **Assumptions:** offer save rates, offer and contact costs, contribution margin, value horizon and discount rate are *visible parameters* with documented sources and tested ranges (Section 8). The brief explicitly allows a synthetic offer-cost/response table; we never treat it as measured.

3 Data Understanding and Preparation

3.1 Source and provenance

The primary dataset is the IBM Telco Customer Churn sample distributed on Kaggle [1]: 7,043 customers of a *fictional* US telco, 21 columns, one row per customer. The pipeline downloads it from Kaggle (with the IBM GitHub copy as a fallback), verifies its SHA-256 checksum (`88be4b93fbe0cc83...`) and refuses to overwrite raw data that does not match. The file was downloaded on 2026-10-03; Kaggle lists version 1, last updated 2018-02-23. A second source, IBM's extended version of the same sample [2], is joined one-to-one on customer ID only to *describe* why customers left; its fields never enter the model (Section 3.4).

Table 2: Data dictionary (grouped). Full dictionary: `docs/data-dictionary.md`.

Group	Fields	Use
Key	customerID	join key only
Demographics	gender, SeniorCitizen, Partner, Dependents	gender and SeniorCitizen: fairness audit only; Partner, Dependents: model inputs
Account	tenure (months), Contract, PaperlessBilling, PaymentMethod	model inputs; tenure also drives the value horizon
Charges	MonthlyCharges, TotalCharges	MonthlyCharges: input and value; TotalCharges dropped (collinear)
Services	PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies	model inputs; eligibility of the service-bundle offer
Target	Churn (Yes/No)	label; 1,869 churners (26.5%)

3.2 Data quality scorecard

The course's data-quality dimensions were checked by an explicit schema in code; any violation stops the pipeline with the row and column that failed (Table 3).

Table 3: Data quality checks on the raw file.

Dimension	Check	Result
Completeness	missing values per column	only `TotalCharges`: 11 blanks
Uniqueness	exact duplicate rows; duplicate IDs	0 duplicates; 7,043 unique IDs
Validity	domains of every categorical field; numbers finite and non-negative; integer tenure	all values valid
Consistency	"No internet service" if and only if InternetService = No; "No phone service" if and only if PhoneService = No	no contradictions
Provenance	source URL, download time, SHA-256, licence note	recorded in `data/raw/source.json`
Cross-source agreement	extended sample: customer ID, churn label and shared fields	7,043 of 7,043 customers matched, all shared fields agree

Cleaning decisions. The 11 blank `TotalCharges` values all belong to customers with `tenure` = 0, i.e. customers who have not yet received a bill. The values are missing for a structural reason (not at random), so they were set to 0 instead of being imputed with a mean. No other rows were filtered. Remaining missing numeric inputs in uploaded files are imputed with *training-split* medians inside the model pipeline, so no information from the validation or test customers leaks into training.

3.3 Validation design

The sample is a single cross-sectional snapshot with no dates, so a temporal hold-out is impossible. We therefore used a stratified random split into train / validation / test = 4,225 / 1,409 / 1,409 customers (60/20/20, churn rate preserved in each part). Every choice (features, hyper-parameters, calibration, model, decision threshold, offer strategy, risk/value cut-offs and the budget) was made on train or validation; the test split was used once for reporting. Because each customer appears exactly once, there is no repeated-entity leakage between splits. With real operator data, the split should become temporal (train on earlier months, test on a later month); this is listed as a deployment requirement.

3.4 Leakage audit

Only information available at the scoring date may be used. Table 4 lists every field that was excluded and why.

Table 4: Leakage and exclusion audit.

Field	Decision and reason
Churn reason, churn category (extended sample)	**Excluded.** Recorded only after a customer has left: target leakage. Used for description only.
IBM churn score, churn value (extended sample)	**Excluded.** IBM's own model output or the label itself.
CLTV index (extended sample)	**Excluded** from the model; used only to cross-check our value proxy.
TotalCharges	**Dropped** by the feature review: it is almost exactly tenure × monthly charges, adds no cross-validated precision and makes coefficients unstable (multicollinearity).
gender, SeniorCitizen	**Excluded** as model inputs on fairness grounds; kept for the allocation audit.
customerID	**Excluded**: an identifier, not a behaviour.

A known limitation of this teaching sample remains: tenure, contract and charges are observed in the same snapshot as the churn flag, so for churners they describe the customer at departure. No feature is derived from the label itself, but with real data the inputs must be taken from a snapshot *before* the outcome window.

4 Exploratory Analysis: Who Churns and Who Is Worth Saving

4.1 Churn drivers

Figure 3 shows observed churn by segment on the training split. Churn is highest for month-to-month contracts (43% against 3% on two-year contracts), new customers with tenure of 12 months or less (48% against 14% for customers with more than 24 months), fibre-optic internet (42% against 7% without internet) and payment by electronic check (46% against 15% for automatic credit-card payment). Customers without TechSupport (42%) or OnlineSecurity (42%) churn more than those with them. Senior citizens churn at 41% against 24%. These are associations in observational data, not causes: moving a customer to a two-year contract will not necessarily remove the risk that led the customer to choose a monthly one.

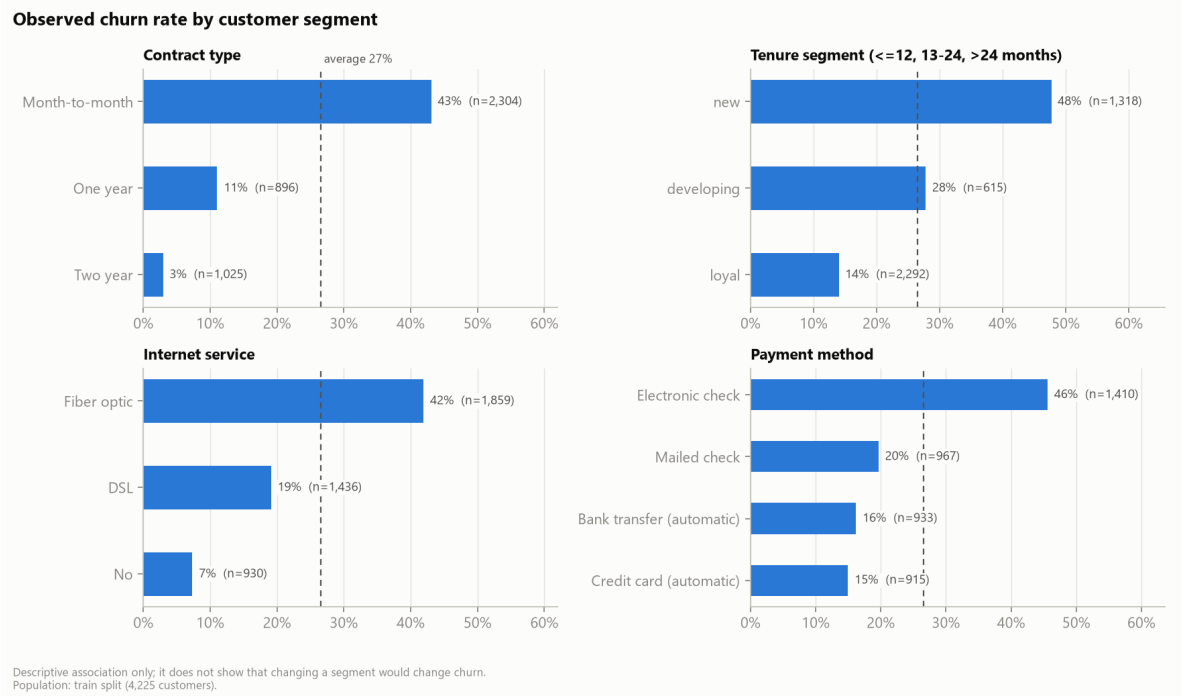


Figure 3: Observed churn rate by segment (training split, 4,225 customers).

4.2 Risk and value only partly overlap

The key business insight of the EDA is that the customers most likely to leave are not the customers most worth keeping. Using the 75th percentiles of predicted risk (0.465) and customer value ($345), fitted on validation and frozen, the test customers fall into four segments (Table 5, Figure 4).

Table 5: Risk × value segments on the test split and how many customers each rule contacts at the base budget.

Segment	Customers	Observed churn	Mean value	Share of value at risk	Contacted: risk-only	Contacted: recommended
High risk / High value	57	54.4%	$402	13.7%	7	56
High risk / Low value	306	59.8%	$221	46.1%	133	85
Low risk / High value	291	19.2%	$414	20.9%	0	12
Low risk / Low value	755	13.8%	$164	19.2%	0	0

The high-risk/low-value segment holds the largest share of value at risk (46%), but each of its customers is worth little; the risk-only rule spends almost its whole budget there. The recommended rule reaches 56 of the 57 high-risk/high-value customers and adds valuable customers at moderate risk.

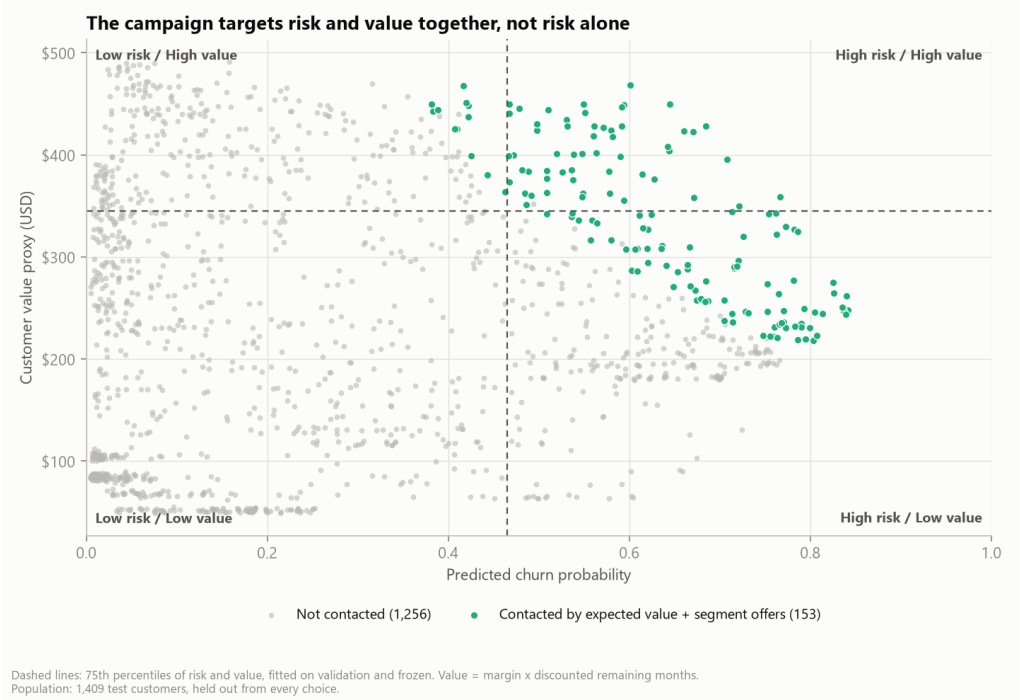


Figure 4: Risk × value map of the test customers. Dashed lines: 75th percentiles fitted on validation.

4.3 Why customers leave

IBM's extended sample records a reason for each of the 1,869 churners. Reasons are known only after departure, so they are used to understand what an offer *can* fix, not to predict (Table 6). Only 46% of churners left for price or a competitor's offer; 42% cite staff attitude or dissatisfaction with the network, product or support, which call for service fixes rather than discounts. This bounds the assumed save rate: if an offer can only reach the addressable churners, a 20% save rate means keeping about 43% of those it can reach, an ambitious figure that must be tested. The most frequent single reason is "Attitude of support person" (10% of churners).

Table 6: Churn reason categories (IBM grouping).

Category	**All** churners	**Churners on the** recommended list	**Addressable by an offer?**
Competitor	33%	33%	Yes
Dissatisfaction	25%	27%	No
Attitude	17%	16%	No
Price	13%	13%	Yes
Other	11%	10%	No

5 Modelling Churn Risk

5.1 Candidates, tuning and selection

Three model families were trained on the same inputs: **logistic regression** (the interpretable baseline), a **random forest** (the required tree-based model) and **XGBoost** gradient boosting (stretch). Each was tuned by 5-fold stratified cross-validation on the training split with average precision (PR-AUC) as the objective, because churners are a minority and the business cares about

the top of the ranking [15]. Each tuned model was then calibrated with a sigmoid (Platt) mapping fitted on out-of-fold predictions [11, 12]. A prior-only model, which predicts the churn rate for everyone, anchors the metrics.

Feature design. Before tuning, a cross-validated review on the training split compared four candidate feature sets. The rule, fixed in advance, was: *full-rank sets only; cross-validated ap within one standard error of the best full-rank candidate; then the fewest encoded columns.* It selected the set "without TotalCharges" (Table 7). Categorical variables are coded against their lowest-churn level (for example, two-year contracts), which avoids the dummy-variable trap and lets odds ratios read "compared with…".

Table 7: Cross-validated feature review (training split, logistic regression).

Feature set	CV PR-AUC	Std. error	Encoded columns	Full rank
all inputs	0.664	0.014	21	yes
without TotalCharges (selected)	0.661	0.013	20	yes
without TotalCharges, with tenure band	0.664	0.013	22	yes
all inputs with tenure band	0.665	0.014	23	yes
previous design (reference)	0.665	0.013	48	no

Selection rule (validation, fixed in advance): *AP within 0.01 of best calibrated candidate; Brier within 0.002 of best; prefer simpler.* Tree models did not rank customers better than logistic regression, so the calibrated **logistic regression** was deployed.

5.2 Performance on the held-out test split

Table 8: Test performance (1,409 customers). Recall is at the F1-optimal threshold chosen on validation (0.40).

Model	PR-AUC	ROC-AUC	Brier	ECE	Cal. slope	Recall
Prior-only baseline	0.265	0.500	0.195	0.022	n/a	0%
Logistic regression	0.620	0.830	0.142	0.021	0.92	65%
Logistic regression, calibrated	0.620	0.830	0.142	0.019	0.94	65%
Random forest	0.621	0.832	0.141	0.020	1.05	62%
Random forest, calibrated	0.621	0.832	0.144	0.045	0.91	59%
XGBoost	0.613	0.826	0.144	0.026	0.84	63%
XGBoost, calibrated	0.613	0.826	0.145	0.044	0.93	57%

The deployed model reaches a PR-AUC of 0.620 (bootstrap 95% interval 0.567–0.671) against a baseline of 0.265, and a ROC-AUC of 0.830 (0.807–0.851). The riskiest decile churns at 71% (lift 2.67), and the top three deciles capture 64% of all churners. Test PR-AUC is lower than on validation (0.672), and the validation value lies outside the test interval: validation was used to choose among candidates, so its score is optimistic, and the test figure is the one to rely on.

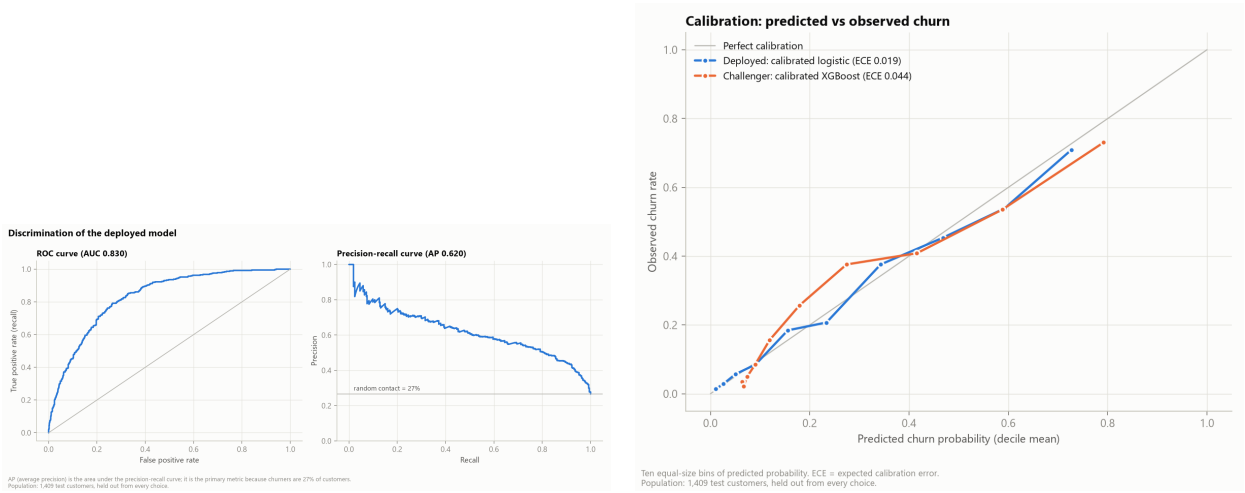


Figure 5: Left: discrimination (ROC and precision–recall) of the deployed model. Right: reliability curve on ten equal-size bins of predicted probability (test split).

Calibration matters as much as ranking. Expected value multiplies a probability by money, so a model that ranks well but over-states risk would over-spend. The deployed model's expected calibration error is 0.019 [13], its calibration slope 0.94 (target 1) and intercept −0.032 (target 0) [14]. Sigmoid calibration made little difference to logistic regression and *worsened* the tree models' ECE on test; we report this as found.

The model choice checked in money. At the base budget, the three calibrated models produce similar expected value on test, but valued with customers' actual churn the logistic list is worth $1,977 against $1,591 for the random forest and $1,468 for XGBoost: the simpler model is not a sacrifice.

5.3 What drives predicted churn

Figure 6 shows the odds ratios of the deployed logistic model. Holding the other inputs fixed, a month-to-month contract multiplies the odds of churn by 3.66 compared with a two-year contract, and fibre internet by 2.90 compared with no internet. Longer tenure lowers the odds (by a factor of 0.69 per 12 months), as do phone service, OnlineSecurity and TechSupport. The XGBoost challenger, explained with TreeSHAP [16], ranks the same drivers first (Contract_Month-to-month, tenure, MonthlyCharges, InternetService_Fiber optic), which increases confidence that the drivers are stable across model families. Both are associations, not causal effects.

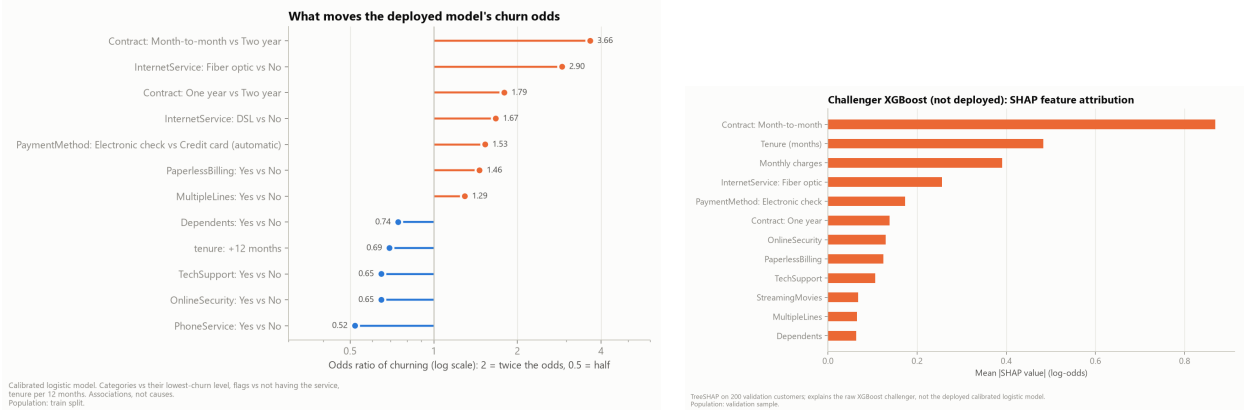


Figure 6: Left: odds ratios of the deployed model (categories against their lowest-churn level, tenure per 12 months). Right: mean absolute SHAP values of the XGBoost challenger on 200 validation customers.

Points scorecard. For the call centre, the logistic model is also published as a points card (base 133 points; every 40 points double the odds of churn). On test the card reproduces the model's probability within 1.0 percentage points on average and has the same ROC-AUC (0.831), so agents can explain a score without software (`docs/churn-scorecard.md`).

Error analysis. At the F1 threshold, false positives (191 customers flagged who stayed) look like true churners: 99% month-to-month and 80% fibre. Missed churners (132) have longer tenure (mean 31 months) and lower bills, i.e. customers whose departure the snapshot does not foreshadow. The weakest segment for ranking is month-to-month contracts (ROC-AUC 0.74), because almost all of them are risky; calibration is weakest for senior citizens (ECE 0.049 against 0.019), which is followed up in the fairness audit.

6 From Prediction to Decision

6.1 Customer value proxy

The data contain no margin or lifetime, so value is a transparent proxy built from tenure and charges, as the brief suggests:

$$V_i \;=\; \text{MonthlyCharges}_i \times m \times a(H_i, r)\,,\;\; H_i = \min\!\big(H_{\max},\, H_0 \cdot f_{\text{seg}(i)}\big),$$
$$a(H, r) = \frac{1 - (1 + r_m)^{-H}}{r_m},$$

with contribution margin $m = 30\%$, base horizon $H_0 = 12$ months scaled by tenure segment ($f = 0.75$ / 1.0 / 1.25 for new / developing / loyal customers), cap $H_{\max} = 24$ months and a 10% annual discount rate (r_m is the equivalent monthly rate). It is a deliberately short, truncated value [10, 9]: a saved would-be churner is likely to leave again sooner than an average customer. As a sanity check, its rank correlation with IBM's CLTV index is only 0.18, because that index is driven by tenure (0.37) whereas our proxy measures the *next* months of margin at stake.

6.2 Expected value of an offer and the decision rule

For customer i and offer o:

$$\text{EV}_{i,o} \;=\; \underbrace{p_i}_{\text{churn risk}} \times \underbrace{q_o}_{\text{save rate}} \times \underbrace{V_i}_{\text{value}} - \underbrace{\big(c_o^{\text{offer}} + c^{\text{contact}}\big)}_{\text{cost}}.$$

The offer cost is charged to every contacted customer (an unconditional credit), which is conservative. A customer is worth contacting only when $\text{EV} > 0$; equivalently, the customer's churn probability must exceed the *break-even probability* $c/(q\,V_i)$, which depends on the customer's value. On test this break-even probability ranges from 29% for the most valuable decile to above 100% (never worth an offer) for the least valuable ones, which is why a single probability threshold cannot be right.

Worked example. A customer pays \$70 a month and has 20 months of tenure ("developing", 12-month horizon). Discounted, the horizon is worth 11.40 months of margin, so V = \$70 × 30% × 11.40 = \$239.41. With $p = 0.60$, the standard offer ($q = 20\%$, cost \$25) gives EV = 0.60 × 0.20 × \$239.41 − \$25 = \$3.73: barely worth it. At this risk the customer would need a value of at least \$208.33 to break even.

Offer catalogue. The brief allows a synthetic, documented offer table. Four offers with eligibility rules were defined (Table 9); their costs and save rates are team assumptions to be replaced by pilot measurements.

Table 9: Synthetic offer catalogue (assumptions, not measurements).

Offer	Offer cost	Contact cost	Save rate	Eligibility
One-time $20 bill credit	$20	$5	20%	All customers
$35 credit for moving to a 1-year contract	$35	$5	30%	Contract: Month-to-month
6 months free TechSupport + OnlineSecurity	$12	$5	15%	InternetService: DSL or Fiber optic; TechSupport: No
$10 loyalty reward	$10	$5	10%	tenure_segment: loyal

6.3 Budget-constrained optimisation

Choosing customers and offers within a budget is a knapsack problem. With binary decision $x_{i,o}$ (customer i receives offer o), budget B, N customers and contact cap κ:

$$\max_{x} \sum_{i,o} \mathrm{EV}_{i,o}\, x_{i,o} \quad \text{s.t.} \quad \sum_{o} x_{i,o} \le 1 \;\; \forall i, \qquad \sum_{i,o} \left(c_o^{\text{offer}} + c^{\text{contact}}\right) x_{i,o} \le B, \qquad \sum_{i,o} x_{i,o} \le \kappa N,$$

with $x_{i,o} = 0$ when the customer is not eligible for the offer or $\mathrm{EV}_{i,o} \le 0$. It is solved exactly as a mixed-integer program (SciPy/HiGHS, zero optimality gap); with a single offer and equal costs, sorting by EV is provably optimal and is used directly. Unit tests check the optimiser against brute force on small cases.

6.4 Threshold rules versus expected value

The baseline brief asks for threshold analysis. The F1-optimal threshold on validation is 0.40, but a threshold ignores value and cost (Table 10): contacting everyone above it costs more and earns less than contacting only customers with positive expected value.

Table 10: Probability threshold versus expected-value rules (standard offer, test split).

Rule	Customers	Cost	Expected net value	Precision	Recall
Probability >= 0.40 (F1 threshold)	433	$10,825	$1,989	55.9%	64.7%
Expected value > 0 (no budget limit)	332	$8,300	$3,202	56.3%	50.0%
Expected value, ranked within the fixed budget	140	$3,500	$2,340	60.0%	n/a

7 Results: Targeting at a Fixed Budget

7.1 Strategy comparison

All rules receive the same budget ($3,500, enough for the standard offer to 10% of the 1,409 test customers). Table 11 and Figure 7 compare them. “Label-based net” revalues each frozen list with customers’ *actual* churn instead of predicted probabilities (still at the assumed save rates).

Table 11: Strategies at the same budget on the test split. HR/HV = high-risk/high-value customers.

Rule	Contacted	Cost	Expected net value	ROI	Precision	HR/HV coverage	Label-based net
No campaign	0	$0	$0	n/a	n/a	0%	$0
Highest churn risk	140	$3,500	$1,301	37%	71%	12%	$1,079
Highest risk, positive EV only	140	$3,500	$1,367	39%	69%	12%	$1,105
Highest customer value	140	$3,500	−$1,037	−30%	29%	30%	$90
Expected value (single offer)	140	$3,500	$2,340	67%	60%	96%	$1,977
Expected value + segment offers	153	$3,500	$2,668	76%	60%	98%	$2,156
Random contact (mean of runs)	140	$3,500	−$1,645	−47%	26%	10%	−$1,688

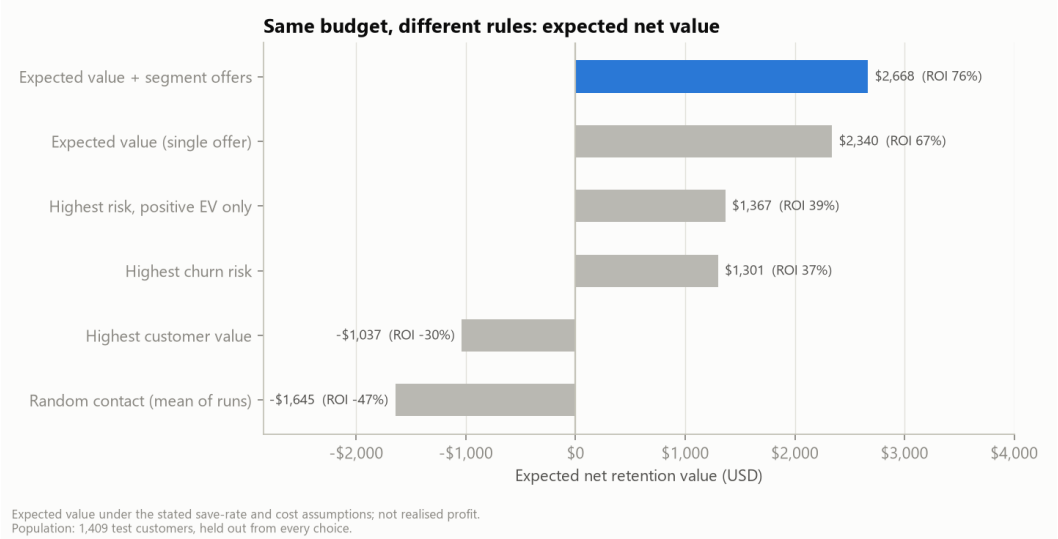


Figure 7: Expected net value of each rule at the same budget (test split). Expected values under the stated assumptions, not realised profit.

- **Targeting effect.** With the same standard offer, ranking by expected value instead of risk adds +$1,039 of expected value; valued with actual churn it adds +$898 (paired bootstrap 95% interval $107 to $1,744), so the gain does not rest on the model's own probabilities.
- **Offer-design effect.** Matching offers to segments adds +$328 expected (label-based +$179, interval −$153 to $577). This part comes from *assumed* offer-specific save rates and is not proven; the pilot tests each offer as its own arm. On validation it beat the single offer by 13%, above the 5% rule fixed in advance, so it is the recommended rule.
- **Precision is not the goal.** Risk-only targeting is more precise (71% vs 60% of contacted customers churn) but spends on risky customers whose remaining value cannot repay an offer. The recommended rule covers 98% of high-risk/high-value customers against 12%.
- **Value alone and random contact lose money**, because offers to customers who would have stayed anyway are pure cost.

7.2 Who receives an offer

On the recommended list, 100% of customers have fibre (44% of all test customers), 97% are on month-to-month contracts (56%) and 73% pay by electronic check (35%); their mean predicted churn is 60% against 27% overall. The optimiser mixes the offers as follows: $35 credit for moving to a 1-year contract (150 customers); 6 months free TechSupport + OnlineSecurity (150 customers);

One-time $20 bill credit (24 customers). The loyalty reward is never chosen under the current assumptions.

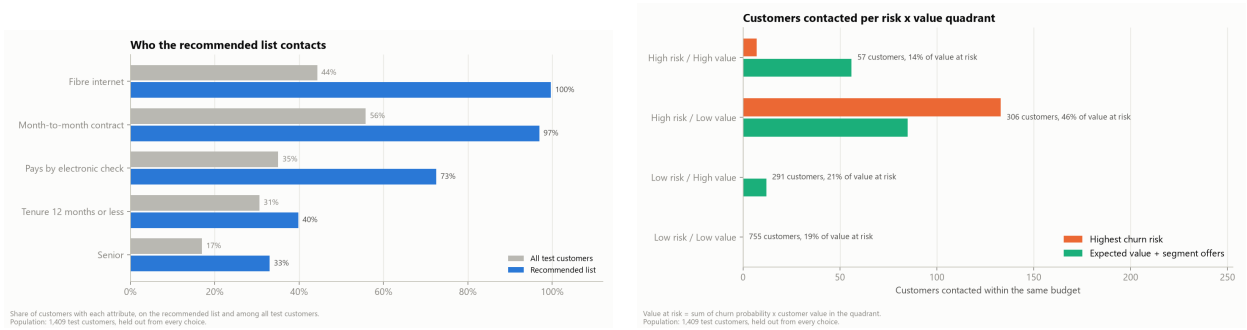


Figure 8: Left: profile of the recommended list against all test customers. Right: which risk/value segments each rule contacts.

7.3 How much to spend: the profit curve

Expected net value rises with the budget at a falling marginal return (Figure 9). The budget rule, fixed before looking at test, is: *take the largest budget whose last step still returns at least 10% on its cost.* Applied on validation it chose 26% (the net-maximising budget on validation was 32%; the hurdle stops earlier because the last steps barely pay for themselves).

Table 12: Base and recommended budgets on the test split.

Budget	Contacted	Cost	Expected net value	ROI	Label-based net
Base (10%)	153 (11%)	$3,500	$2,668	76%	$2,156
Recommended (26%)	324 (23%)	$9,150	$4,145	45%	$3,377

Beyond 371 contacts on test (26% of customers) no customer has a positive expected value, so extra budget should stay unspent. Scaled linearly to the 7,043-customer base, the recommended wave is about 1,620 customers and $45,737 for $20,721 of expected net value (indicative only).

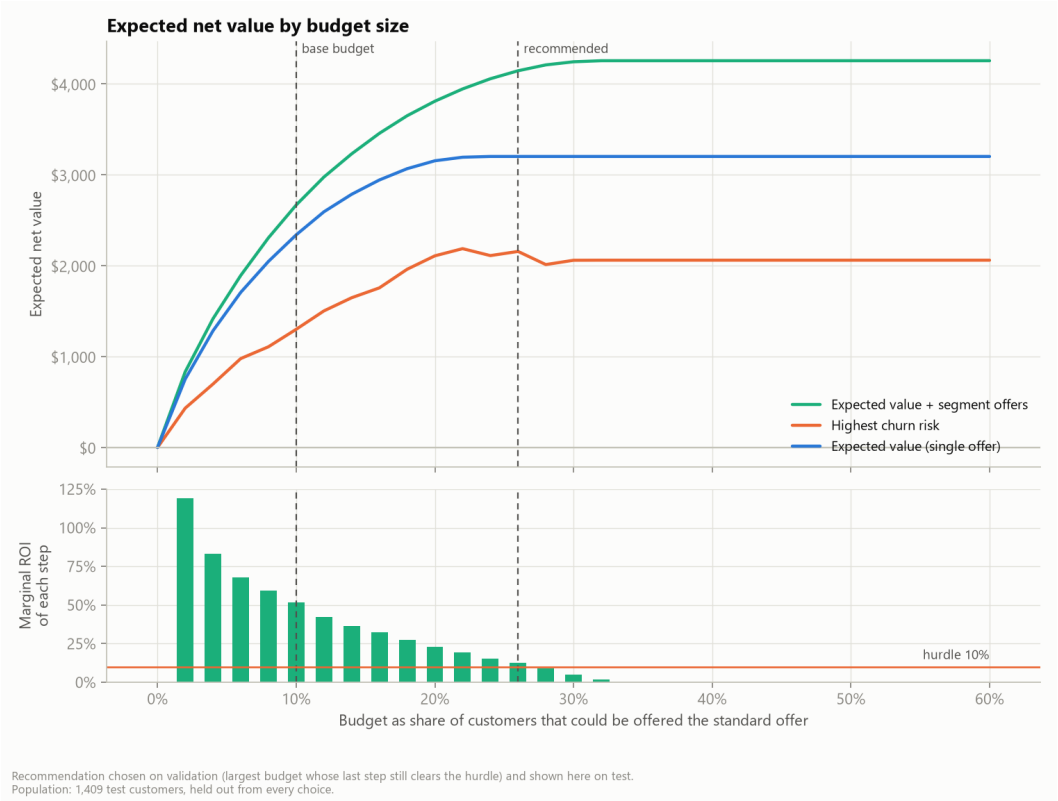


Figure 9: Profit curve and marginal ROI by budget (test split). The recommendation was chosen on validation.

8 Robustness and Sensitivity

Every money figure depends on assumptions the data cannot measure. They are listed with sources and tested ranges in the assumption register (Table 13) and are adjustable in the dashboard.

Table 13: Assumption register (base value and tested range).

ID	Assumption	Low	Base	High	Validate by
A1	Save rate of the standard offer	10%	20%	30%	Randomised holdout pilot (docs/pilot-design.md)
A2	Offer cost (multiplier on catalog cost)	x0.5	x1	x1.5	Finance sign-off of the offer budget
A3	Contact cost per customer (USD)	$1	$5	$10	Call-centre cost per completed contact
A4	Contribution margin	20%	30%	40%	Finance (product P&L)
A5	Value horizon (months)	6 months	12 months	18 months	Survival of pilot customers after 6 and 12 months
A6	Annual discount rate	5%	10%	15%	Finance hurdle rate
A7	Tenure effect on remaining months (new 1-s, loyal 1+s)	0%	25%	50%	Survival of pilot customers by tenure segment

8.1 One assumption at a time

Each bar of the tornado chart (Figure 10) moves one assumption to the end of its range. Two versions are reported: a *re-optimised* plan, which re-selects customers knowing the new value, and

a *frozen* list, which keeps the customers already chosen, as happens once a list has been sent.

Table 14: One-way sensitivity of the expected net value (base budget, test split).

Assumption	Low	High	Net at low	Net at high	Frozen at low	Frozen at high
Save rate of the standard offer	10%	30%	$458	$6,359	−$469	$5,806
Value horizon (months)	6 months	18 months	$89	$5,522	−$336	$5,522
Offer cost (multiplier on catalog cost)	x0.5	x1.5	$6,061	$1,198	$4,036	$1,301
Contribution margin	20%	40%	$614	$4,724	$612	$4,724
Contact cost per customer (USD)	$1	$10	$3,843	$1,852	$3,280	$1,903
Annual discount rate	5%	15%	$2,836	$2,513	$2,836	$2,513
Tenure effect on remaining months (new 1-s, loyal 1+s)	0%	50%	$2,973	$3,280	$2,427	$2,885

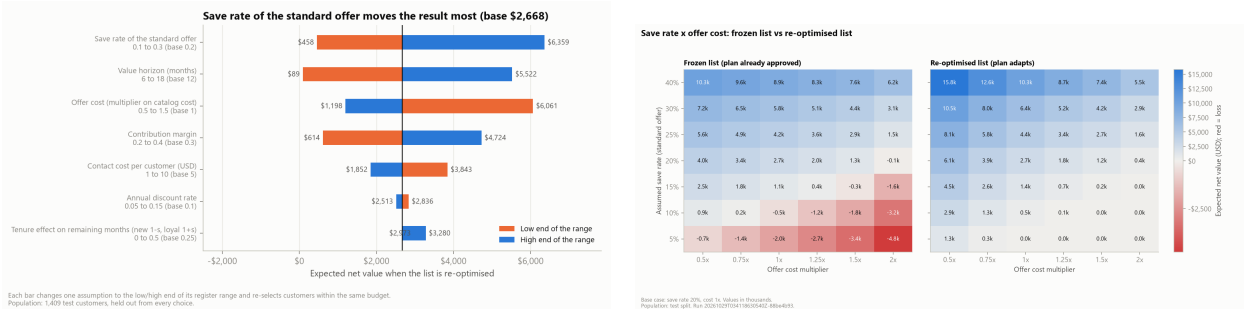


Figure 10: Left: tornado chart. Right: save rate × offer cost grid, frozen and re-optimised lists.

Over the save-rate × offer-cost grid the re-optimised plan ranges from $0 to $15,784, whereas a frozen list ranges from −$4,773 to $10,311. A re-optimised plan can never lose money only because it pretends the true assumptions are known; a list that is actually sent can.

Break-even. The base-budget list breaks even at a standard-offer save rate of **11.3%** or at offer costs 1.98× today's; the larger recommended list breaks even at **13.8%**, because its additional customers have lower expected value. Both figures scale every offer's save rate in proportion and are expressed as the standard offer's rate; measured as the average save rate of the recommended offer mix, which is what a pilot observes, the break-even is 15.6% against an assumed 22.6%. Below a standard-offer save rate of 7.7% no customer-offer pair has a positive EV and the right decision is not to run a campaign.

Table 15: Combined scenarios (base budget).

Scenario	Assumptions	Contacted	Net (re-optimised)	Net (frozen)
Pessimistic	save rate 10%, offer cost x1.25, contact cost $8, margin 25%, value horizon 9 months	0	$0	−$2,724
Base	assumptions as registered	153	$2,668	$2,668
Optimistic	save rate 30%, offer cost x0.75, contact cost $2, margin 40%, value horizon 15 months	312	$19,084	$12,958

8.2 All assumptions uncertain at once

The Monte Carlo analysis draws every register assumption at once from a triangular distribution over its range (mode at the base value), 5,000 times, and reprices the frozen lists (Table 16, Figure 11). A frozen list is repriced at the drawn costs, so when offers turn out dearer it spends more than its budget, as a list already sent would; the re-optimised comparison always stays within the budget.

Table 16: Monte Carlo of all assumptions (5,000 draws, frozen lists, test split).

List	Mean net	5th pct.	Median	95th pct.	P(loss)
Base-budget list	\$2,593	−\$469	\$2,375	\$6,282	9%
Recommended list	\$3,997	−\$2,444	\$3,639	\$11,532	17%
Risk-only list	\$1,241	−\$1,267	\$1,080	\$4,304	24%

- The recommended list earns more on average but is more exposed: it loses money in 17% of draws against 9% for the smaller base-budget list.
- Against the risk-only list at the same budget, expected-value targeting earns more in 99.96% of draws (median gain \$1,240): **who to target is robust; how much an offer saves is not.**
- Value horizon (months), save rate of the standard offer and contribution margin drive most of the spread (rank correlations 0.59, 0.51, 0.42).
- Knowing the true assumptions before sending a list would be worth about \$846 per campaign of this size (\$4,227 across the base). This is the value of the information a pilot buys, and it recurs with every later wave.

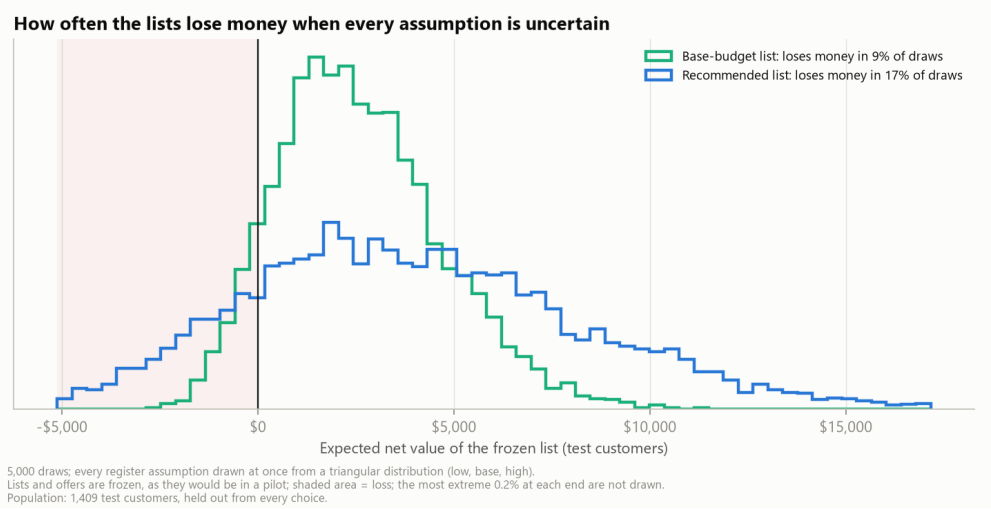


Figure 11: Distribution of net value when all assumptions vary at once. Shaded area: loss.

8.3 Outside the sample: lower churn and Vietnamese prices

The sample's 27% churn is far above that of real operators (US carriers report about 1% monthly postpaid churn [21, 22]). To show what that does to the answer, all predicted probabilities were shifted by one constant on the log-odds scale until the average churn fell to 15%, 10%, 5%, a prior-shift adjustment in the spirit of [17], with the constant chosen so that the average equals the target; the ranking of customers is unchanged. A second set of rows prices the same customers

like a Vietnamese home-fibre operator (a 250,000 VND bill, about $9.76; offers scaled to the same share of the bill; contact by one SMS of 500 VND).

Table 17: Stress test without a budget limit. Worth an offer: customers with at least one positive-EV offer; expected net: the campaign under the contact cap.

Prices	Average churn	Worth an offer	Share	Expected net (USD)	Pilot size per arm
Sample company (USD)	27%	371	26%	$4,256	287
Sample company (USD)	15%	105	7%	$519	447
Sample company (USD)	10%	25	2%	$63	523
Sample company (USD)	5%	0	0%	$0	n/a
Vietnam-like prices	27%	463	33%	$948	304
Vietnam-like prices	15%	247	18%	$205	525
Vietnam-like prices	10%	98	7%	$48	689
Vietnam-like prices	5%	5	0%	$0	916

Lower churn shrinks the target list fast: at 15% average churn only 7% of customers are worth the standard offer (against 26% in the sample). With Vietnamese prices the money per customer is small, but an SMS costs almost nothing, so 18% of customers remain worth an offer at 15% churn: *cheap contact matters more than the offer*. The pilot must also grow, from 287 customers per arm to 447 at 15% churn.

9 Fairness, Ethics and Customer Experience

Gender and age are not model inputs, but they are used to audit *who receives offers* [18]. Selection-rate ratios outside 0.8–1.25 (the "four-fifths" screening analogy [19]) are flagged for human review rather than corrected automatically (Figure 12).

- **Senior citizens** are contacted 3.34× as often as other customers under the recommended rule (2.64× under risk-only targeting). Adjusted for risk decile the ratio falls to 1.65×: higher churn risk explains part of the gap and higher bills (value) the rest. Being selected means receiving both a discount and a call, so the gap is a benefit and a burden at once; it is flagged for review by Legal/DPO in every wave.
- **Gender:** ratio 1.15× (95% interval 0.85–1.58), within normal variation.
- **Loyalty penalty:** long-tenure and contract customers are rarely contacted because their risk is low. To avoid the perception that only those who threaten to leave are rewarded [20], the playbook recommends a non-targeted loyalty benefit funded outside the retention budget.
- **Contact burden and compliance:** at most 30% of customers per wave, one retention contact per 30 days, consent and do-not-call checks; in Vietnam, advertising calls are limited to 08:00–17:00 and one per subscriber per 24 hours [23], and personal data processing requires purpose-specific consent [24].

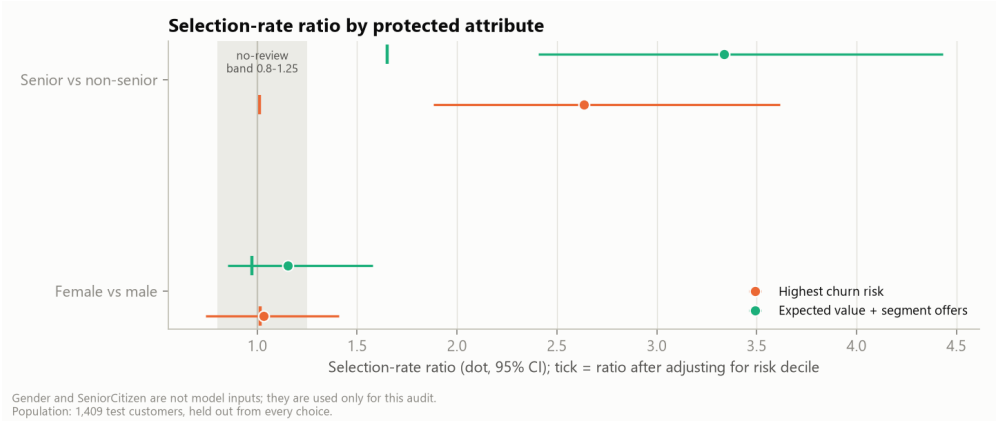


Figure 12: Selection rate by group relative to the reference group, for each targeting rule.

10 Deployment: Dashboard and Pilot

10.1 Management dashboard

A Streamlit dashboard (`app/Home.py`) turns the analysis into a decision tool, following the course's dashboard principles: one question per page, the key numbers top-left, colour used only for meaning. The *Decision* page (Figure 13) answers "which customers should get an offer?" with five KPIs: expected net value (with the gap to risk-only), ROI, customers contacted, precision among contacted and high-risk/high-value coverage. It also shows a one-sentence recommendation, who is and is not on the list, and an exportable action list with the reason for every decision. Budget and offer policy are inputs; the *Assumptions & sensitivity* page exposes every assumption, the break-even save rate, the live chance of a loss and the stress test; the *Model evidence* page shows performance, the scorecard and the SHAP challenger, and scores uploaded customer files. The default view reproduces the numbers in this report exactly (same scenario ID).

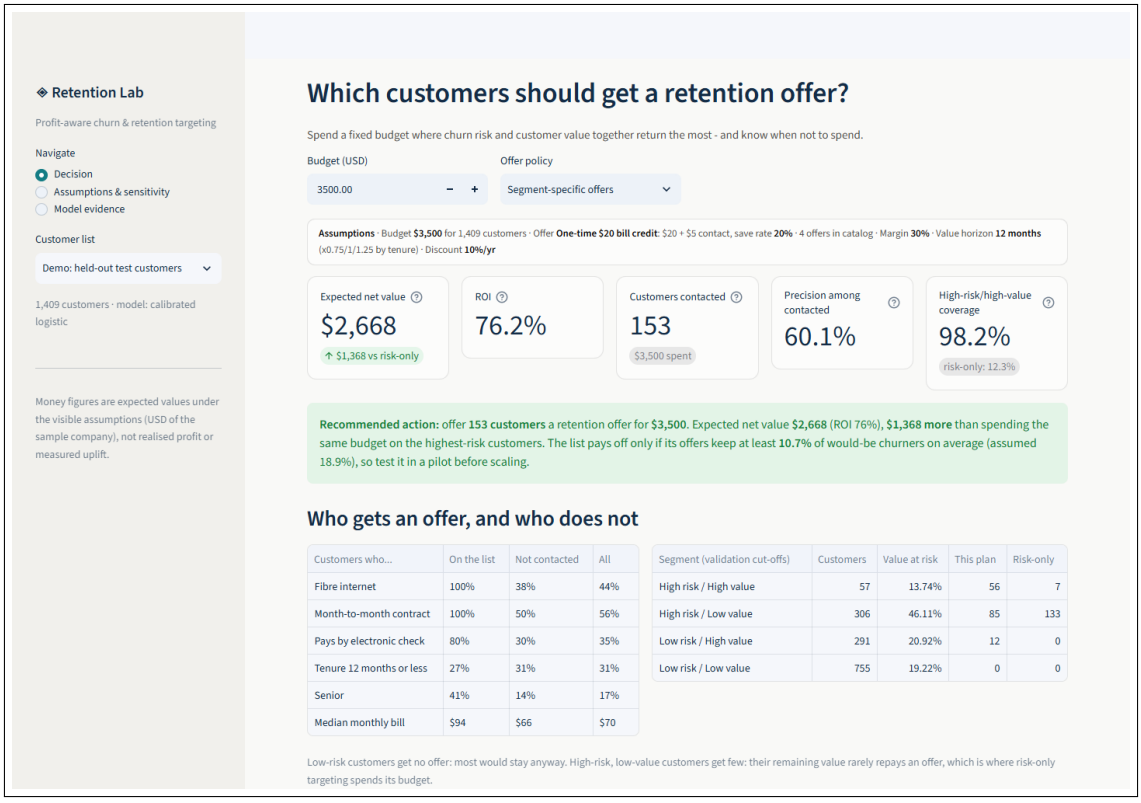


Figure 13: Decision page of the dashboard (default assumptions, held-out test customers).

10.2 Randomised pilot

Because the save rate cannot be estimated from these data, the recommendation is deployed as an experiment [8, 5]. One wave of 1,620 customers selected by the recommended rule is split at random: 810 receive the offer and 810 are held out. The save rate is the relative reduction of churn in the offered half against the held-out half after 90 days. Because the offered half receives a mix of offers, this measures the mix's average save rate, assumed to be 22.6%; the list breaks even at 15.6% on this scale.

- **Power.** With 5% significance and 80% power, detecting the assumed 22.6% save rate needs 213 customers per arm, and detecting the break-even rate (15.6%) needs 447; one wave of 810 per arm covers both.
- **Decision rule, agreed in advance:** go if the 95% interval of q is above 0 and q >= break-even; stop if q <= 0 or the interval is below break-even; otherwise extend (q = 1 - churn(offered) / churn(holdout), the list's mix save rate); at most 2 extra waves. If the true save rate is the assumed 22.6%, the rule says "go" with probability 97%.
- **Cost and downside:** offer and contact cost about $22,875 for the offered half, expected net value $10,363, worst case (nobody saved) −$22,875. A decision is available about 19 weeks after approval.

Table 18: Roadmap from approval to scale-up.

Phase	Weeks	Owner	Work
Set-up	4	Retention, Legal/DPO, IT	consent and Do-Not-Call checks, data-protection impact assessment, CRM list export, call script and agent training
Pilot wave	13	Call centre, Retention	contact the offered half once; follow both halves for the 90-day outcome window
Read-out	2	Retention analytics	save rate with its interval, net value per customer, guardrail metrics; go, extend or stop
Scale or stop	4	Head of Retention, Finance	retrain and recalibrate on own data, replace assumed save rates, roll out the recommended budget

11 Limitations and External Validity

1. **No causal uplift.** The data contain no randomised campaign, so save rates are assumptions and no uplift is claimed. High-risk customers may be the least responsive, and proactive contact can even raise churn [6, 5]. The pilot is designed to measure this; after several waves, uplift models can replace the save-rate assumption [7].
2. **Teaching sample.** A fictional US company with 27% churn, far above real operators; IBM describes the churn window inconsistently (month vs quarter). The results demonstrate a method; their magnitudes do not transfer.
3. **Snapshot and random split.** No dates, hence no temporal validation and no drift estimate; features are observed in the same snapshot as the label. Real deployment requires a time-based split and monitoring.
4. **Value proxy.** No cost-to-serve, no re-churn after a save, no network effects; margin and horizon are assumptions and drive most of the uncertainty.
5. **Label-based checks** reuse the assumed save rate. They validate *who* is targeted, not *whether offers work*.
6. **Vietnamese context.** About 90% of Vietnamese mobile subscriptions are prepaid [25], so churn appears as inactivity or number porting and contract features do not exist; the model must be rebuilt on operator data.

12 Conclusion and Recommendations

Risk alone is the wrong criterion for spending a retention budget. A calibrated churn model, a transparent value proxy and an explicit expected-value rule select customers whose saved margin repays the offer, and at the same budget this delivers +$1,368 (105%) more expected value than calling the riskiest customers. The ranking advantage is robust to every assumption we tested; the absolute payoff is not, because it depends on a save rate nobody has measured.

Recommendations to the Head of Retention:

1. Adopt expected-value targeting with segment offers for the next wave, at the recommended budget ($6.49 per customer in the base).
2. Run it as a randomised pilot of 1,620 customers, half held out, 90-day outcome; go only if the effect is significant and the save rate is at least 15.6%.

3. Replace the assumed save rates, margin and costs with measured values and re-run the pipeline; retrain on the operator's own data with a time-based split.
4. Log customers' reasons for leaving and act on service problems (42% of churners cite attitude or dissatisfaction), which no offer budget fixes.
5. Operate under the guardrails: contact cap, consent, fairness review of every wave, and a non-targeted loyalty benefit.

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A Reproducibility and Project Package

Table 19: Where each required component of the project package is.

Required component	Location in the repository
Reproducible code and notebooks, README	`src/telco_retention/`, `notebooks/` (four notebooks), `README.md`
Data dictionary and source log	`docs/data-dictionary.md`, `docs/source-log.md`, `docs/cleaning-log.md`
Executive brief (max. 2 pages)	`reports/executive-brief.pdf`
Interactive dashboard	`app/Home.py` (Streamlit)
Model / analysis card	`docs/model-card.md`
Churn scorecard and targeting table	`reports/tables/scorecard.csv`, `reports/tables/targeting_recommended.csv`, `docs/churn-scorecard.md`
Retention playbook	`docs/retention-playbook.md`, `docs/pilot-design.md`
Assumptions with sources	`configs/default.yaml`, `docs/assumption-register.md`
This report, the presentation and its script	`reports/final-report.pdf`, `reports/presentation/retention-presentation.pptx`, `reports/presentation/speaker-script.md`

How to reproduce. With Python 3.13:

```
python -m venv .venv
.venv\Scripts\python -m pip install -r requirements.lock.txt
.venv\Scripts\python -m pip install -no-deps -e .
.venv\Scripts\python -m telco_retention run-all
.venv\Scripts\python -m streamlit run app/Home.py
```

Each run is stored under `artifacts/runs/<run id>` with its configuration, data checksum, dependency versions and code hashes. The automated test suite (`pytest`) covers data validation, the model, the decision engine (including the optimiser against brute force), the analysis, the reports and the dashboard; a release script verifies that two independent runs produce identical metrics, customer lists and scenario identifiers.

B Final Submission Checklist

Table 20: Course checklist and where it is satisfied.

Checklist item	Evidence
Business decision, unit of analysis, target and evaluation period defined	Section 2, Table 1
Source, download date, licence and filtering/joining documented	Section 3.1–3.2, `docs/source-log.md`
No target leakage; predictors available at decision time	Section 3.4, Table 4
Baseline complete and compared fairly with advanced models	Section 5, Tables 7–8
Validation matches the structure of the problem	Section 3.4 (snapshot data: stratified split, test used once)
Technical metrics and a business metric reported	Sections 5 and 7
Scenario assumptions visible, with sensitivity analysis	Section 8
Dashboard supports a concrete decision	Section 10.1
Executive brief states recommendation, impact range, risks, limitations	`reports/executive-brief.pdf`; Executive Summary
A third party can rerun the analysis	Appendix A

C Team Contributions

Table 21: Contributions of the members of Group 1.

Member	Student ID	Main contributions
Pham Tuan Kiet	20252751M	Code review and reproducibility check; final report and presentation slides
Nguyen Trung Hieu	20261059M	Decision engine, robustness and Monte Carlo analysis, dashboard redesign, business documents
Bui Tien Dung	20261204M	Review and quality assurance of code, report and slides
Pham Hung Phong	20261061M	Data pipeline, churn models and first version of the dashboard